

**Teorema 1.** Suppose  $\mathbb{E}(\exp(i\xi\Gamma)) = \exp(im\xi - \frac{\sigma^2\xi^2}{2})$ . Then  $m = \mathbb{E}(\Gamma)$  and  $\sigma^2 = \text{Var}(\Gamma)$ .

*Proof.* We only examine the case for  $m$ , since the case for  $\sigma^2$  is similar. Observe that  $\mathbb{E}(\frac{d}{d\xi}\exp(i\xi\Gamma)) = \frac{d}{d\xi}\mathbb{E}\exp(i\xi\Gamma)$  (why??). Evaluating both sides we get:

$$\mathbb{E}(i\Gamma \cdot \exp(i\xi\Gamma)) = (im - \sigma^2\xi) \cdot \exp(im\xi - \frac{\sigma^2\xi^2}{2})$$

Evaluating in  $\xi = 0$  we obtain:

$$i \cdot \mathbb{E}(\Gamma) = i \cdot m$$

□